

Pranav Gunreddy

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EDUCATION

University of Pennsylvania, U.S.A

Master of Science and Engineering in Mechatronics & Robotics

Aug'22 – May'24

Indian Institute of Technology Jammu, India

Bachelor of Technology in Mechanical Engineering

Aug'17 – Jun'21

TECHNICAL SKILLS

Languages: Embedded C, C++, Python | **frameworks:** CMSIS, TinyML, pyTorch, HuggingFace, GTSAM, GYM, NeuroMancer
Tools: Git, CMake, Docker | **Hardware:** ARM Cortex-M, Teensy, Jetson Nano, Arduino, RPi | Moteus, CAN, SPI, UART
Concepts: Scientific deep learning, reinforcement learning, sensor fusion, signal processing, Embedded ML

EXPERIENCE

R&D Control Systems & Algorithms Engineer | SharkNinja - Boston, Massachusetts | Jun '24 - *

Embedded system design, thermal control, UART, C++, Matlab, Stateflow, Python, CAD, PCB design

- Designed and deployed real-time control & sensor fusion algorithms for complex smart appliances, operating under strict compute, power, latency, and cost constraints. (multiple patent families for coffee, slushed drink, smart cooking machines)
- Designed embedded architectures with adaptive signal processing techniques for robust low-latency control
- Explored optimization model deployment strategies for real-time inference on Cortex-M-class embedded systems

R&D Software Engineer | Solidworks Assemblies, Dassault Systemes | Jun '21 - Jul '22

C++, API design, Data structures, Object oriented Design, MFC, STL, Computational geometry

- Used Computational Geometry methods to implement new CAD features and enhanced existing software modules

R&D Control Software Intern | Verdant Robotics - Hayward, California | May'23 - Aug '23

C++, Docker, CI/CD, PID, Autotuning, Sys ID, GTSAM, CAN, Imitation Learning, RL, NARX, NODE

- Improved weed hit rate at high vehicle speeds through adaptive, learning based control & uncertainty modeling
- Used Autoregressive models, Neural ODEs and RL to setup real time robot health monitoring and turret-crop assignment
- Worked on HIL Testbed, modified sensor drivers, resolved CAN protocol, latency, time sync & telemetry issues
- Cascaded, FOC based control with Moteus firmware to drive 2 phase BLDC motors, Simulink Coder (C++ code generation)

Research Assistant - Learning based control & Safe planning | GRASP Lab | Sept '22 - Apr '23

CARLA, PINN, DeepONet, FNO, Conformal predictions, LSTM, (Robust, Hybrid, Explicit) MPC

- Explored Physics informed Deep learning to learn Non linear dynamics & dynamic pedestrian trajectory predictions
- Incorporated safety guarantees and created Robust, Hybrid, Explicit MPC frameworks with these learnt models

PROJECTS

Grand Theft Autonomous - Mobile Battlebot | 🌐

Itsy Bitsy, ESP32, SPI, I2C, UDP, Wifi communication, Vive, Infrared, PID

- Designed & manufactured a battle bot capable of autonomous maneuvering. Built filtering circuits, localization, wireless comm modules. Used low cost motor drivers & sensors, power regulators, comm protocols for a cheap yet robust & safe design

BUD - Navigation assist Mobile Robot, SICK LiDAR Challenge | 🌐

LiDAR, IMU, Docker, ROS, Gazebo, Move Base, hector SLAM, Dynamometer

- Built mobile robot with an interactive force control interface to autonomously assist people to navigate through indoor spaces
- Performed extensive market study and business analysis to produce a low cost, working prototype

Underactuated Robotics & Optimization | 🌐

Dynamic Programming, MPC, iLQR, Reinforcement learning, OSC, Drake, casadi, GYM

- Implemented advanced control for underactuated systems (cartpole, pendulum) using Dynamic programming, iLQR & RL
- Implemented Operational space controller for planar biped robot with High DOF

Pick & place in dynamic environment using 7 DOF Panda arm | 🌐

ROS, Gazebo, FK, IK, A, RRT*, APF, Bezier curve interpolation, Software design, System Integration*

- Implemented FK/IK/Trajectory generation for a 7-DOF arm to perform static, dynamic pick & place while avoiding obstacles
- Integrated perception and state estimation with position and velocity control module